BUSITEMA UNIVERSITY AILAB/NLP INTERNSHIP PROGRAM

Week 1-Day 2: Machine Translation

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1. Introduction

Machine Translation is a Natural Language Processing task that involves translating text from one language to another. The objective of this task was to implement and compare three different translation approaches.

2. Dataset

The dataset used was an English–French parallel corpus sourced from Kaggle. It consisted of two files containing aligned sentences in English and their corresponding French translations.

Preprocessing:

- Combined English and French datasets
- Removed missing values
- Split into training and validation sets

3. Models

3.1 Baseline Model

A simple dictionary-based translation approach was used, mapping English sentences directly to French equivalents.

3.2 Seq2Seq Model

A sequence-to-sequence model using LSTM was implemented to learn translation patterns from the dataset.

3.3 Pretrained Model

A pretrained MarianMT transformer model was used for high-quality translation.

4. Evaluation

BLEU score was used to evaluate translation quality by comparing predicted translations with reference translations.

5. Results

Model	**BLEU Score**
Baseline	0.00
Seq2Seq	0.00(greater than 0)
MarianMT	0.49

6. Discussion

The Baseline model achieved a BLEU score of 0.00 because it relies on direct sentence matching and cannot generalize to unseen inputs.

The Seq2Seq model also produced a BLEU score close to zero, indicating poor performance. This is due to the complexity of machine translation tasks and the limited training configuration used.

The MarianMT model significantly outperformed the other models, achieving a BLEU score of approximately 0.49. This is because it is a pretrained transformer model trained on large multilingual datasets, enabling it to generate accurate and context-aware translations.

These results highlight the effectiveness of pretrained transformer models compared to traditional and from-scratch approaches.

7. Conclusion

Transformer-based models significantly outperform traditional approaches in machine translation tasks. The MarianMT model provided the most accurate translations.